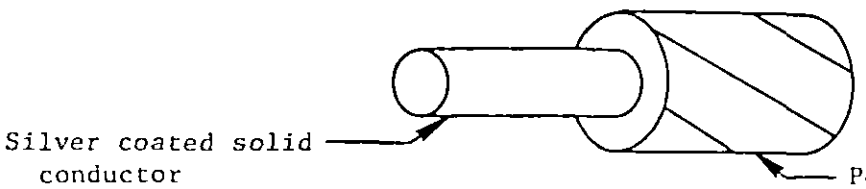
MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, SOLDERLESS WRAP, POLYETHYLENE TEREPHTHALATE - POLYESTER LAMINATED TAPE INSULATION, SILVER COATED SOLID CONDUCTOR

Ⓐ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the wire described herein shall consist of this specification and the issue in effect of Specification MIL-W-81822.



Ⓐ

TABLE I. Construction and performance details.

Part no. 1/	Wire size (AWG)	Finished wire diameter (inches)	Finished weight Ⓐ (lb/1,000 ft) (max)	Cut-through load (grams)	Bend testing			Insulation pull-off (lbs)
					Mandrel diameter (ins.)		Tension load (lbs)	
					Heat resistance	Cold bend	Cold bend	
M81822/1-X30-*	30	0.0195 ± .0010	0.48	1400	1/16	1.0	0.25	0.4 to 2.5 Ⓐ
Ⓐ M81822/1-X30V*	30	0.0180 ± .0010	0.45	1400	1/16	1.0	0.25	0.4 to 2.5
M81822/1-X28-*	28	0.0265 ± .0015	0.83	1800	5/64	1.0	0.50	0.5 to 2.5
Ⓐ M81822/1-X28V*	28	0.0255 ± .0015	0.80	1800	5/64	1.0	0.50	0.5 to 2.5
M81822/1-X26-*	26	0.0295 ± .0015	1.17	2200	3/32	1.0	0.50	1 to 4
Ⓐ M81822/1-X26V*	26	0.0265 ± .0015	1.08	2200	3/32	1.0	0.50	1 to 4
M81822/1-X24-*	24	0.0340 ± .0015	1.73	2600	7/64	1.0	0.50	1 to 4
M81822/1-X22-*	22	0.0390 ± .0020	2.58	2800	1/8	1.0	0.75	1 to 6
M81822/1-X20-*	20	0.0460 ± .0020	3.92	2900	9/64	1.0	0.75	1 to 6
M81822/1-X18-*	18	0.0540 ± .0030	6.00	3000	11/64	1.0	1.00	1 to 6

1/ PART NUMBER: The "X" in the generalized part numbers of table I shall be replaced by the designator for the conductor type and the asterisk shall be replaced by the code designator for the insulation color. Example: Size 24 wire with type A (soft annealed copper) conductor and white insulation - M81822/1-A24-9.

Ⓐ 2/ Polyvinyl chloride laminant is not permitted.

Ⓐ Denotes changes.

FSC 6145

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WIRE RATINGS AND ADDITIONAL REQUIREMENTS

Ⓐ TEMPERATURE RATING: 105°C (221°F) max conductor temperature.
VOLTAGE RATING: 300 volts (rms).
INSULATION TENSILE STRENGTH: 2,500 psi (min).
INSULATION ELONGATION: Wire sizes 30 and 28 - 35 percent (min).
Wire sizes 26 through 18 - 50 percent (min).
COLOR: MIL-STD-104, class 1, color as specified in contract or order.
(All MIL-STD-104, class 1, colors are available.)
Ⓐ SPARK TEST: 2,000 volts (rms).
INSULATION RESISTANCE: 500 megohms-1,000 ft (min).
Ⓐ WET DIELECTRIC TEST: 1,000 volts (rms), 60 Hz.
COLD BEND CONDITIONING: 4 hrs at -54° ± 2°C (-65.2° ± 3.6°F).
DIELECTRIC TEST FOR CUT-THROUGH: 500 volts (rms), 60 Hz.
Ⓐ INSULATION HEAT RESISTANCE:
Conditioning, 48 hrs at 150° ± 2°C (302° ± 3.6°F).
Shrinkage, 0.250 inch (max).
DIELECTRIC CONSTANT: 4.00 max at 1 MHz.
POWER FACTOR: 0.05 max at 1 MHz.
Ⓐ RESISTANCE TO FLUIDS, APPLICABLE TEST FLUIDS:
Isopropyl alcohol, TT-I-735.
Hydraulic fluid, MIL-H-5606.
Trichlorotrifluoroethane, MIL-C-81302.
Trichloroethane, MIL-T-81533.
FLAMMABILITY: 30 sec (max) after-flame.
3.0 inches (max) flame travel.
No flaming of tissue paper.
SURFACE RESISTANCE: 100 megohms (min).
Ⓐ FUNGUS RESISTANCE: Wire constructed as specified in MIL-W-81822/1A shall be considered fungus resistant in accordance with Requirement 4 of MIL-STD-454 by virtue of the recognized inherent fungus resistance of all components. Fungus resistance test is not required.

Custodians:
Army - CR
Navy - AS
Air Force - 95

Preparing activity:
Navy - AS

(Project 6145-0683-2)

Review activities:
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Air Force - 11, 99
DLA - ES, IS

User activities:
Navy - MC, OS
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DOCUMENT IDENTIFIER (Number) AND TITLE MIL-W-81822/1A WIRE ELECTRICAL, SOLDERLESS WRAP, POLYETHYLENE TEREPHTHALATE - POLYESTER LAMINATED TAPE INSULATION, SILVER COATED SOLID CONDUCTOR

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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